

Business Case Study – Aerofit Product Analysis

Colab Link -

https://drive.google.com/file/d/1Ky2zo7xjY_gAwWm2HaUdlCxjvD7ZjyDq/view

1. Executive Summary

This case study analyzes 180 customer purchase records from Aerofit to understand customer behavior across three treadmill products: KP281 (Entry Level), KP481 (Mid Level), and KP781 (Premium Level).

The objective was to identify customer demographics driving each product, income and fitness influence on purchase, usage behavior patterns, probability of product purchase, and key correlations impacting premium product selection.

The analysis reveals that income, fitness level, and usage frequency are the strongest drivers of premium product purchase, while younger and lower-income customers prefer entry-level products.

2. Business Objective

1. Identify customer segments for each product tier.
2. Understand factors influencing premium product purchase.
3. Analyze impact of income on product selection.
4. Build customer profiles for targeted marketing.

3. Data Understanding

Total Records: 180

Products: KP281, KP481, KP781

Features: Age, Gender, Education, Marital Status, Usage, Fitness, Income, Miles

Engineered Features:

- Product Tier (Entry / Mid / Premium)
- Price Mapping
- Age Groups
- Usage Level Buckets
- Fitness Level Buckets

4. Data Cleaning & Preparation

- No missing values observed.
- Income values rounded to integers.
- Outliers identified using IQR method.
- Extreme values clipped between 5th and 95th percentile.

Outlier Observations:

Income: 19

Miles: 13

Price: 40 (due to tier pricing structure)

5. Key Analytical Insights

Income vs Product Tier:

Clear upward trend — higher income customers prefer Premium products.

Gender Insights:

Female customers strongly prefer Entry model.

Male customers show significant preference for Premium.

Age Group Insights:

17–25 → Mostly Entry

26–35 → Entry + Mid

36+ → Higher Premium inclination

Fitness & Usage:

High fitness and high weekly usage customers prefer Premium products.

6. Probability Analysis

Overall Marginal Probability:

KP281 – 44.4%

KP481 – 33.3%

KP781 – 22.2%

Gender-Based Probability:

Male customers show higher probability of purchasing Premium.

Female customers show higher probability of purchasing Entry model.

7. Correlation Insights

Strong positive correlations observed:

Usage & Miles → Heavy users run more.

Fitness & Miles → Fit users run longer distances.

Income & Price → Higher income leads to higher tier purchase.

Usage & Fitness → Frequent usage improves fitness.

Premium buyers typically:

- High income
- High usage
- High fitness
- Higher miles covered

8. Customer Profiling

KP281 – Entry Level Customer:

Age: 17–30

Income: Low to Mid

Fitness: Medium

Usage: 2–3 times/week

Mostly Female customers

KP481 – Mid Level Customer:

Age: 26–35

Income: Mid-range

Fitness: Medium to High

Usage: 3–4 times/week

Often Partnered

KP781 – Premium Customer:

Age: 30+

Income: High

Fitness: High

Usage: 4–5 times/week

Predominantly Male

9. Business Recommendations

1. Target Premium marketing to high-income, high-fitness customers.
2. Promote Entry model to young working professionals.
3. Create upgrade journey from Entry → Mid → Premium.
4. Offer EMI options to increase Premium adoption.
5. Use behavior-based segmentation instead of only demographic targeting.

10. Strategic Conclusion

Income, fitness level, and weekly usage are the strongest predictors of premium purchase.

By implementing segmentation-driven marketing strategies, Aerofit can increase premium conversion rates, improve customer lifetime value (CLV), and enhance marketing ROI.